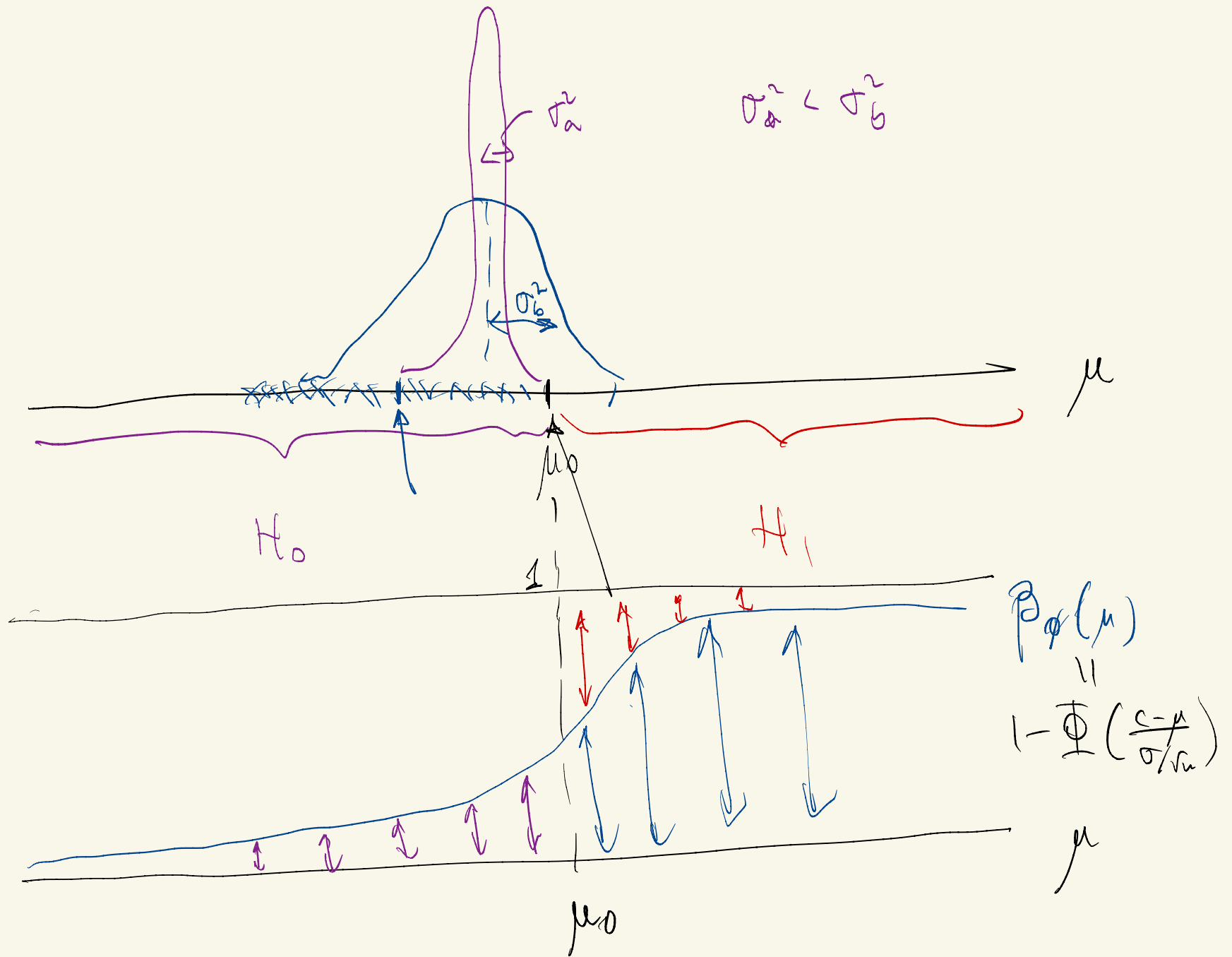


ORF 524 - Fall 2026
Week Q5



$$H_0: \mu \leq \mu_0$$

vs

$$H_1: \mu > \mu_0$$

Power

$$X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\beta_\phi(\mu) = \mathbb{P}_\mu [\bar{X}_n > c] = \mathbb{P} [\text{rejecting } H_0 \mid \mu \text{ distributed}]$$

$$H_0: \mu \leq \mu_0 \quad \text{vs} \quad H_1: \mu > \mu_0$$

$$\beta(\mu) = \mathbb{P}_\mu [\bar{X}_n > c]$$

$$= \mathbb{P}_{\cancel{X}} \left[\underbrace{\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}}_{\mathcal{N}(0,1)} > \frac{c - \mu}{\sigma/\sqrt{n}} \right]$$

$$= 1 - \Phi\left(\frac{c - \mu}{\sigma/\sqrt{n}}\right); \quad \Phi(u) = \int_{-\infty}^u \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}v^2} dv$$

We want an α -level test:

$$\sup_{\mu \leq \mu_0} \underbrace{\left(1 - \Phi \left(\frac{c - \mu}{\sigma/\sqrt{n}} \right) \right)}_{P_{\mu} [\bar{X}_n > c]} \leq \alpha$$

$$\Rightarrow 1 - \Phi \left(\frac{c - \mu_0}{\sigma/\sqrt{n}} \right) = \alpha$$

$$\Rightarrow c = \mu_0 + \Phi^{-1}(1 - \alpha) \cdot \frac{\sigma}{\sqrt{n}}$$

Neyman-Pearson lemma

$$H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta = \theta_1$$

MP (simple-vs-simple) test:

$$\Lambda(x; \theta_0, \theta_1) = \frac{f(x; \theta_1)}{f(x; \theta_0)} > k \quad \Leftrightarrow \quad \text{Reject } H_0.$$

$$\mathbb{E}_{\theta_0} \left[\mathbb{1}(\Lambda(x, \theta_0, \theta_1) > k) \right] = \alpha$$

$$= \mathbb{P}_{\theta_0} \left[\Lambda(x; \theta_0, \theta_1) > k \right]$$

Proof: Recall: Reject $H_0 \Leftrightarrow \phi^* = 1 \Leftrightarrow \frac{f_1(x)}{f_0(x)} > k$

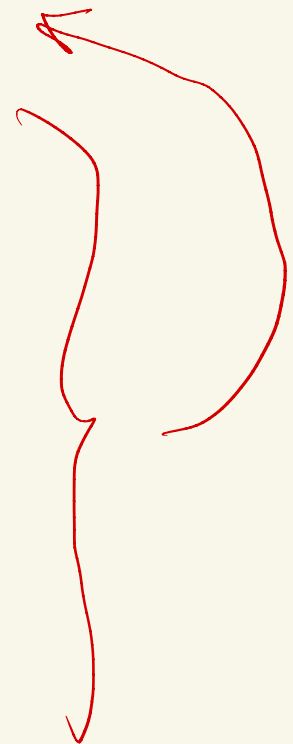
$$(\phi^* - \phi)(f_1 - k f_0) \geq 0$$

$$(1 - 1) \cdot \quad = 0$$

$$(1 - 0) \cdot \quad \textcircled{+} \quad > 0$$

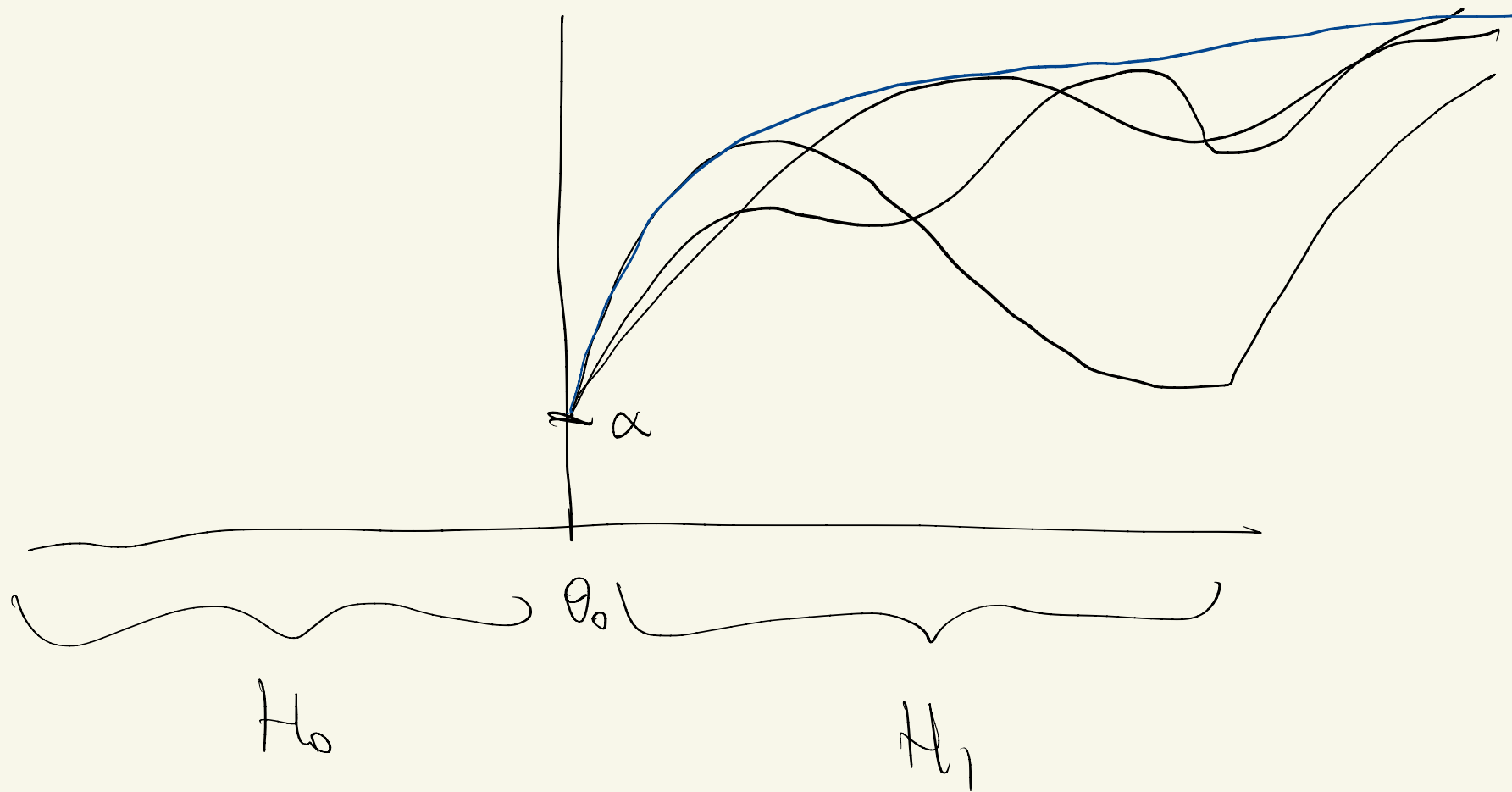
$$(0 - 1) \cdot \quad \textcircled{-} \quad > 0$$

$$(0 - 0) \cdot \quad = 0$$



$$\Rightarrow \int (\phi^*(x) - \phi(x)) \cdot (f_1(x) - k f_0(x)) dx \geq 0$$

$$\Rightarrow \int \phi^*(x) \cdot f_1(x) dx - \int \phi(x) f_1(x) dx - k \underbrace{\int (\phi^*(x) - \phi(x)) f_0(x) dx}_{= 0} \geq 0$$



$\boxed{\text{blue}} = \text{UMP}$

$$H_0: \theta = \theta_0$$

vs.

$$H_1: \theta \neq \theta_0$$

